



ABBOTSLEIGH

Student Number: _____

AUGUST 2010

YEAR 12

ASSESSMENT 4

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks (120)

- Attempt Questions 1-10.
- All questions are of equal value.

Outcomes assessed

Preliminary course

- P1 demonstrates confidence in using mathematics to obtain realistic solutions to problems.
- P2 provides reasoning to support conclusions that are appropriate to the context
- P3 performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4 chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5 understands the concept of a function and the relationship between a function and its graph
- P6 relates the derivative of a function to the slope of its graph
- P7 determines the derivative of a function through routine application of the rules of differentiation
- P8 understands and uses the language and notation of calculus

HSC course

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts
- H2 constructs arguments to prove and justify results
- H3 manipulates algebraic expressions involving logarithmic and exponential functions
- H4 expresses practical problems in mathematical terms based on simple given models
- H5 applies appropriate techniques from the study of calculus, geometry, trigonometry and series to solve problems
- H6 uses the derivative to determine the features of the graph of a function
- H7 uses the features of a graph to deduce information about the derivative
- H8 uses techniques of integration to calculate areas and volumes
- H9 communicates using mathematical language, notation, diagrams and graphs

Question 1 (12 Marks)

Marks

- (a) Evaluate $\log_e 1.6$ correct to three decimal places. 2
- (b) Solve $5 - 6x \leq 3$ 2
- (c) Factorise fully: $24x^3 + 3y^3$ 2
- (d) Simplify $\frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta}$ 2
- (e) Find the equation of the line passing through the point $(-1, 2)$ and parallel to $3x - y + 4 = 0$ 2
- (f) Find the limiting sum of the geometric series $\frac{3}{2} + 1 + \frac{2}{3} + \dots$ 2

Question 2 (12 Marks)

Start a new booklet

Marks

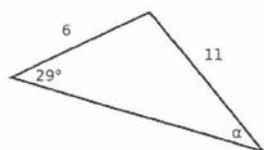
- (a) Solve $\sqrt{3} \tan \theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$ 3
- (b) Differentiate with respect to x :
- (i) $\frac{5x}{x^2 - 3}$ 2
- (ii) $(1 + \cos x)^5$ 2
- (c) (i) Find $\int 1 + e^{5x} dx$ 2
- (ii) Evaluate $\int_0^{\frac{\pi}{6}} \sin 2x dx$ 3

Question 3 (12 Marks)

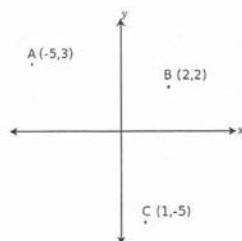
Start a new booklet

Marks

- (a) Find the value of α in the diagram (Diagram is not to scale). Give your answer to the nearest degree. 2



- (b) In the diagram A , B and C are the points $(-5, 3)$, $(2, 2)$ and $(1, -5)$ respectively (Diagram is not to scale).



- (i) Calculate the gradient of AC . 1
- (ii) Find the coordinates of X , the midpoint of AC . 1
- (iii) Calculate the length of BX . 1
- (iv) Hence, or otherwise, find the coordinates of D if X is the midpoint of BD . 1
- (v) Show that $AC \perp BD$. 2
- (vi) Explain why the quadrilateral $ABCD$ is a square. 2
- (vii) Calculate the area of $ABCD$. 2

Question 4 (12 Marks)

Start a new booklet

Marks

- (a) Find the equation of the normal to the curve $f(x) = 6 \ln(x - 1)$ at the point $(2, 0)$. 3

- (b) The number of seats in each row of a theatre increases by 4 as you go from the front row to the back row.

- (i) If there are fifteen seats in the front row, show that there are $(4n+11)$ seats in the n th row. 1

- (ii) If the theatre has 18 rows of seats, calculate the total number of seats in the theatre. 2

- (c) A survey shows that if Australian voters were asked to choose who they preferred as Prime Minister, Julia Gillard or Tony Abbott, 45% would choose Julia Gillard, 40% would choose Tony Abbott and the remainder would be undecided.

If two voters were chosen at random and asked to make this choice:

- (i) Find the probability that they would both be undecided. 1
- (ii) Find the probability that at least one would be undecided. 2

- (d) The pendulum arm on a grandfather clock is 1 metre in length.

- (i) If it sweeps out a sector of area $\frac{\pi}{12}$ square metres as it swings, show that the angle through which the pendulum arm moves is $\frac{\pi}{6}$. 1

- (ii) How far apart are the end points of the arm at the two extremes of the pendulum swing? 2

Question 5 (12 Marks)

Start a new booklet

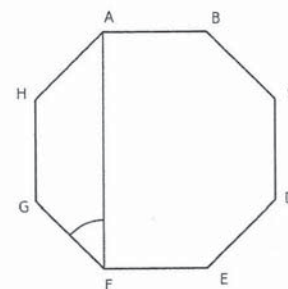
Marks

- (a) The roots of the quadratic equation $px^2 - x + q = 0$ are -1 and 3 . Find p and q . 2
- (b) A function is defined by $g(x) = (x^2 - 4)(2 - x)$
- (i) Find all solutions of $g(x) = 0$ 2
- (ii) Find the coordinates of the turning points of the graph of $y = g(x)$, and determine their nature. 3
- (iii) Sketch the graph of $y = g(x)$ showing the turning points and the x and y intercepts. 2
- (iv) For what values of x is the graph concave up? 1
- (c) Use Simpson's rule with three function values to find an approximation to the value of $\int_0^2 \sqrt{4 - x^2} dx$. Give your answer correct to three significant figures. 2

Question 6 (12 Marks)

Start a new booklet

Marks



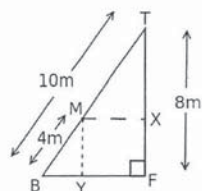
- (a) $ABCDEFGH$ is a regular octagon (Diagram is not to scale). Find the size of $\angle AFG$ giving all reasons. 3
- (b) Find the coordinates of the focus of the parabola $x^2 = -12(y - 3)$. 2
- (c) Sunlight transmitted into water loses intensity as it penetrates to greater depths. Intensity I at depth d metres below the surface is given by $I = Ae^{-kd}$. If the intensity of sunlight 300 metres below the surface of the water is three-tenths of the original intensity find:
- (i) the value of k 2
- (ii) the depth below the surface (to the nearest metre) at which the intensity would be halved. 2
- (d) Solve for x : 3
- $$3e^{-2x} - e^{-x} - 2 = 0$$

Question 7 (12 Marks)

Start a new booklet

Marks

- (a) A ladder TB , 10 metres long, rests against a vertical wall with the top, T , 8 metres up the wall (Diagram is not to scale).

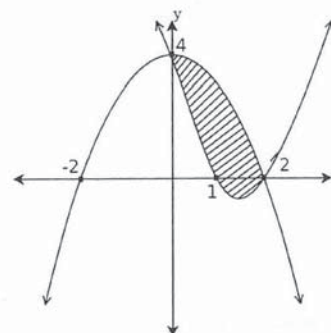


- (i) How far is the base of the ladder, B , from the foot of the wall, F ? 1
- (ii) Milli stands at the point M , 4 metres from the base of the ladder, with Y vertically below her on the ground and X horizontally across from her on the wall. Prove that $\triangle MYB$ is similar to $\triangle TFB$. 2
- (iii) How far above ground level is Milli? 2
- (b) Merrilyn, a graduate engineer, earns \$60 000 in her first year of employment and in each of the following years her annual salary is increased by 5% of the previous year's salary.
- (i) What is Merrilyn's annual salary in her 5th year of employment? 2
- (ii) Calculate her total earnings for the first ten years of her employment. 2
- (iii) At the end of each year of employment Merrilyn invests \$5 000. Her investment earns interest at a rate of 4% per annum, compounded annually. Calculate the total amount she has accrued by the end of her first 10 years of employment. 3

Question 8 (12 Marks)

Start a new booklet

Marks



- (a) The shaded region is bounded by two parabolae. (Diagram is not to scale) The parabola $y = 4 - x^2$ cuts the x axis at $(-2, 0)$ and $(2, 0)$ and the y axis at $(0, 4)$. The other parabola cuts the x axis at $(1, 0)$ and $(2, 0)$ and the y axis at $(0, 4)$.
- (i) State the roots of the other parabola. 1
- (ii) Hence, or otherwise, show that the equation of the other parabola is $y = 2x^2 - 6x + 4$. 1
- (iii) Use calculus to find the area of the shaded region. 3
- (b) $\int_0^1 \frac{e^x}{1 + e^x} dx = \log_e c$. Find c . 3
- (c) Consider the equation in x : $(2k - 3)x^2 + (k + 1)x - 1 = 0$
- (i) For what value of k is this a linear equation? 1
- (ii) Find all values of k for which this is a quadratic equation with two distinct roots. 3

Question 9 (12 Marks)

Start a new booklet

Marks

(a) (i) Draw a neat sketch of the curve $y = \sec \frac{x}{2}$ for $-2\pi \leq x \leq 2\pi$ 2

(ii) A section of the curve $y = \sec \frac{x}{2}$ from $x = -\frac{\pi}{2}$ to $x = \frac{\pi}{2}$ is rotated about the x axis to form a solid of revolution. Calculate the exact volume of this solid. 3

(b) The gradient of a curve at any point is given by $f'(x) = e^{-2x}$.

(i) Explain why the curve $y = f(x)$ is an increasing function for all values of x . 1

(ii) Given that the curve has a y intercept of 4.5, use $f'(x)$ to show that

$$f(x) = 5 - \frac{e^{-2x}}{2}$$

2

(iii) What is $\lim_{x \rightarrow \infty} f(x)$? 1

(iv) Calculate the exact value of the x intercept of the curve. 2

(v) Draw a neat sketch of $f(x)$, showing the asymptote. 1

Question 10 (12 Marks)

Start a new booklet

Marks

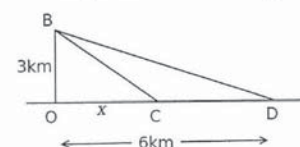
(a) $A(4, 2)$ and $B(-2, -8)$ are two points on the number plane. The point $P(x, y)$ moves so that PA is always perpendicular to PB .

(i) Find the gradients of PA and PB in terms of x and y . 1

(ii) Hence, show that the equation of the locus of P is $(x-1)^2 + (y+3)^2 = 34$ 3

(iii) Describe the locus of P geometrically. 1

(b) A man in a boat at B is 3km from the nearest point O on a straight beach (Diagram is not to scale). He wants to get to his beachhouse at D , 6 kilometres along the beach from O .



(i) He decides to row to a point C , between O and D , and then walk from C to D . If C is x kilometres from O , find an expression for the distance BC he will row. 1

(ii) If he can row at 4 km/h and walk at 5 km/h, show that the total time in hours, $T(x)$, he will take to row to C then walk to D is given by

$$T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6-x}{5} \text{ for } 0 \leq x \leq 6$$

2

(iii) Find the value of x for which the total time he will take to get to his beach house is a minimum. 3

(iv) What is the least amount of time he will take to reach his beach house? 1

END OF PAPER

1(a) $\log_e 1.6 = 0.47000 \dots$
 $\div 0.470$

(b) $5 - 6x \leq 3$
 $-6x \leq -2$

$x \geq \frac{1}{3}$
 (c) $24x^3 + 3y^3 = 3(8x^3 + y^3)$
 $= 3(2x+y)(4x^2 - 2xy + y^2)$

(d) $\frac{\sin^2 \theta + \cos^2 \theta}{\tan^2 \theta} = \frac{1}{\tan^2 \theta}$
 $= \cot^2 \theta$

(e) par. to $3x - y + 4 = 0$
 $y = 3x + 4$
 grad $m = 3$, $(-1, 2)$ on line

$y - y_1 = m(x - x_1)$
 $y - 2 = 3(x + 1)$
 $y - 2 = 3x + 3$

$3x - y + 5 = 0$

(f) $\frac{2}{3} + 1 + \frac{2}{3} + \dots$
 geom $a = \frac{2}{3}$, $r = \frac{2}{3}$

$S_{\infty} = \frac{a}{1-r}$
 $= \frac{\frac{2}{3}}{1-\frac{2}{3}}$
 $= \frac{\frac{2}{3}}{\frac{1}{3}}$
 $= \frac{2}{1}$
 $= \frac{2}{1}$

2(a) $\sqrt{3} \tan \theta + 1 = 0$, $0 \leq \theta \leq 2\pi$
 $\tan \theta = -\frac{1}{\sqrt{3}}$
 $\theta = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$

$= \frac{5\pi}{6}$ or $\frac{11\pi}{6}$

(b)(i) $\frac{d}{dx} \left(\frac{5x}{x^2-3} \right) = \frac{(x^2-3) \cdot 5 - 5x \cdot 2x}{(x^2-3)^2}$

$= \frac{5x^2 - 15 - 10x^2}{(x^2-3)^2}$

$= \frac{-5x^2 - 15}{(x^2-3)^2}$

$= \frac{-5(x^2+3)}{(x^2-3)^2}$

(ii) $\frac{d}{dx} (1 + \cos x)^5 = 5(1 + \cos x)^4 \cdot -\sin x$
 $= -5 \sin x (1 + \cos x)^4$

(c)(i) $\int (1 + e^{5x}) dx = x + \frac{1}{5} e^{5x} + C$

(ii) $\int_0^{\frac{\pi}{6}} \sin 2x dx = \left[-\frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{6}}$
 $= \left[-\frac{1}{2} \cos \frac{\pi}{3} + \frac{1}{2} \cos 0 \right]$
 $= -\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 1$
 $= \frac{1}{4}$

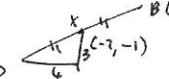
3(a)  $\frac{\sin d}{6} = \frac{\sin 29^\circ}{11}$
 $\sin d = \frac{6 \sin 29^\circ}{11}$
 $= 0.2644 \dots$
 $d = 15.33 \dots$
 $= 15^\circ$ (n. deg).

(b)(i) $A(-5, 3)$ $B(2, 2)$ $C(1, -5)$

$m_{AC} = \frac{y_2 - y_1}{x_2 - x_1}$
 $= \frac{-5 - 3}{1 + 5}$
 $= \frac{-8}{6}$
 $= -\frac{4}{3}$

(ii) $X = \left(\frac{-5+1}{2}, \frac{3-5}{2} \right)$
 $= \left(-\frac{4}{2}, -\frac{2}{2} \right)$
 $= (-2, -1)$

(iii) $BC = \sqrt{(2+2)^2 + (2+1)^2}$
 $= \sqrt{16+9}$
 $= 5$

(iv) $D(-6, -4)$ 

(v) $m_{BD} = \frac{2+4}{2+6}$
 $= \frac{6}{8}$
 $= \frac{3}{4}$
 so $m_{BD} \times m_{AC} = \frac{3}{4} \times -\frac{4}{3}$
 $= -1$

$\therefore BD \perp AC$

(vi) It is a rhombus as diagonals bisect at 90° . $m_{AB} = \frac{2-3}{2+5} = -\frac{1}{7}$ $m_{BC} = \frac{2+5}{2-1} = 7$

so $m_{AB} \times m_{BC} = -\frac{1}{7} \times 7$
 $= -1$

so $AB \perp BC$

so square as adjacent sides right $\angle$ between them.

3(b)(ii) equal diagonals
 since square.
 $A = \frac{1}{2}xy$ (rhombus)
 $= \frac{1}{2} \times 10 \times 10$
 $= 50$
 so 50 units²

4(a) $f(x) = 6 \ln(x-1)$

$f'(x) = \frac{6}{x-1}$

grad of tangent at $(2, 0)$

$f'(2) = \frac{6}{2-1}$
 $= 6$

$\therefore$ incl grad $-\frac{1}{6}$

eq'n of line $y-0 = -\frac{1}{6}(x-2)$

$y = -\frac{1}{6}(x-2)$

$6y = -x+2$

$x+6y-2=0$

(b)(i) 15, 19, 23, ...

arith. $a=15, d=4$

$T_n = 15 + (n-1) \times 4$

$= 15 + 4n - 4$

$= 11 + 4n$

So $(4n+11)$ seats in n th row

(ii) 18 rows $S_{18} = \frac{18}{2}(2 \times 15 + (18-1) \times 4)$

$= 9(30 + 68)$

$= 882$

There are 882 seats.

(c)(i) $(0.15)^2 = 0.0225$

(or 2.25%)

(ii) $P(\text{at least 1 under}) = 1 - P(\text{neither under})$

$= 1 - (0.85)^2$

$= 0.2775$

(or 27.75%)

(d)(i) $A = \frac{1}{2} r^2 \theta$, $r=1$, $A = \frac{\pi}{12}$

$\frac{\pi}{12} = \frac{1}{2} \theta$

$\theta = \frac{\pi}{6}$ so $\frac{\pi}{6}$ rad.

(ii) $d^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos \frac{\pi}{6}$

$= 2 - 2 \times \frac{\sqrt{3}}{2}$

$= 2 - \sqrt{3}$ so $(2 - \sqrt{3})$ m apart.

5(a) $px^2 - x + q = 0$ roots $-1, 3$

$p+1+q=0$ ①
 $q-3+q=0$ ②

② ① $8p-4=0$

$p = \frac{1}{2}$

sub in ① $\frac{1}{2} + 1 + q = 0$

$q = -\frac{3}{2}$

$\therefore p = \frac{1}{2}, q = -\frac{3}{2}$

or $-\frac{b}{a} = \alpha + \beta, \frac{c}{a} = \alpha\beta$

$\frac{1}{p} = -1+3, \frac{q}{p} = -3$

$\frac{1}{p} = 2, 2q = -3$

$p = \frac{1}{2}, q = -\frac{3}{2}$

(b) $g(x) = (x^2-4)(2-x)$

(i) $g(x) = 0 \quad (x^2-4)(2-x) = 0$

$(x-2)(x+2)(2-x) = 0$

$\therefore x = 2 \text{ or } -2 \text{ or } 2$

so $x = 2 \text{ or } -2$

(ii) $g(x) = 2x^2 - x^3 - 8 + 4x$

$g'(x) = 4x - 3x^2 + 4$

$g''(x) = 4 - 6x$

pts $g'(x) = 4x - 3x^2 + 4 = 0$

$3x^2 - 4x - 4 = 0$

$(3x+2)(x-2) = 0$

$x = -\frac{2}{3} \text{ or } 2$

$g(-\frac{2}{3}) = -\frac{256}{27}, g(2) = 0$

$(-\frac{2}{3}, -\frac{256}{27}), (2, 0)$

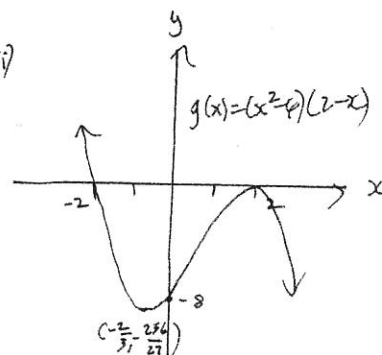
nature? $g'(-\frac{2}{3}) = 4 - 6(-\frac{2}{3}) = 4 + 4 = 8 > 0$

$g'(2) = 4 - 6(2) = -8 < 0$

so conc up ∇ so conc down $\ntriangledown$

min at $(-\frac{2}{3}, -\frac{256}{27})$ max at $(2, 0)$

5(b)(iii)



(iv) $g''(x) > 0$

$4 - 6x > 0$

$-6x > -4$

$x < \frac{2}{3}$

So conc up for $x < \frac{2}{3}$

(c) $\int_0^2 \sqrt{4-x^2} dx$ $\frac{x}{2} \sqrt{4-x^2} + \frac{1}{2} \int \frac{1}{\sqrt{4-x^2}} dx$

$= \frac{1}{6}(y_0 + 4y_1 + y_2)$

$= \frac{2}{6}(2 + 4\sqrt{3} + 0)$

$= 2.9760...$

$= 2.976 \text{ (3 dp)}$

6(a) Octagon. int sum $(n-2) \times 180^\circ$

$= 6 \times 180^\circ$

$= 1080^\circ$

each < of reg oct = $\frac{1080^\circ}{8}$

$= 135^\circ$

AFGH is a quad so < sum 360°

isos trapezium as $AF = GH$

(reg. octagon)

$\therefore \angle AFG = \frac{360^\circ - 2 \times 135^\circ}{2}$

$= 45^\circ$

($\angle HAF = \angle AFG$ in isos. trap.)

(b) $x^2 = -12(y-3)$ $x^2 = 4a(y-k)$

vertex $(0, 3)$

focal length $a = 3$

$\therefore$ focus $(0, 0)$

(c)(i) $I = A e^{-kt}$ orig. intensity is $I = A$

when $d = 300$, $I = 0.3A$

so $0.3A = A e^{-300k}$

$0.3 = e^{-300k}$

$-300k = \ln 0.3$

$k = -\frac{\ln 0.3}{300}$

(ii) Solve $I = 0.5A$

$0.5A = A e^{-\frac{d \ln 0.3}{300}}$

$\frac{d \ln 0.3}{300} = \ln 0.5$

$d = \frac{300 \ln 0.5}{\ln 0.3}$

$= 172.714...$

so approx 173 m below.

$$6d) 3e^{-2x} - e^{-x} - 2 = 0$$

$$\text{let } u = e^{-x}$$

$$3u^2 - u - 2 = 0$$

$$(3u+2)(u-1) = 0$$

$$u = -\frac{2}{3} \text{ or } 1$$

$$e^{-x} = -\frac{2}{3} \text{ or } e^{-x} = 1$$

$$\text{As } e^{-x} > 0 \quad x = 0$$

$$\therefore x = 0$$

$$7a) (i) 6m \text{ (Pythagorean)}$$

$$(ii) \triangle MYB \text{ and } \triangle TXM$$

$$\angle MYB = \angle TXM = 90^\circ$$

$$(\angle MYLB, \angle TXFLBF)$$

$$\angle MBY = \angle TMX \text{ (corresponding angles)}$$

$$\therefore \triangle MYB \cong \triangle TXM \text{ (AAS)}$$

$$(iii) \frac{MB}{TB} = \frac{MY}{TF} \text{ (corresponding sides in } \triangle MYB \text{ and } \triangle TXM)$$

$$\frac{4}{10} = \frac{MY}{8}$$

$$MY = \frac{8 \times 4}{10}$$

$$= 3.2$$

So 3.2m above ground

$$(bxi) T_5 = 60000 \times 1.05^4$$

$$= 72930.375 \dots$$

$$\text{So approx } \$72930.38$$

$$(ii) S_{10} = \frac{60000(1.05^{10} - 1)}{1.05 - 1}$$

$$= 754673.552 \dots$$

$$\text{Approx } \$754673.55$$

$$(iii) \text{At end of 1st yr has } A_1$$

$$A_1 = 5000$$

$$\text{and } A_2 = 1.04 \times 5000 + 5000$$

$$A_3 = 1.04^2 \times 5000 + 1.04 \times 5000 + 5000$$

$$A_n \text{ geom. sum } a = 5000, r = 1.04$$

end of 10th yr

$$A_{10} = \frac{5000(1.04^{10} - 1)}{1.04 - 1}$$

$$= 60030.535 \dots$$

$$\text{So approx } \$60030.54$$

$$8a) (i) 1, 2$$

$$(ii) \alpha + \beta = 3 = -\frac{b}{a} \quad \alpha\beta = 2 = \frac{c}{a} \text{ etc}$$

$$\text{or } y = a(x-1)(x-2)$$

$$\text{Sub in } (0, 4) \quad 4 = a(-1)(-2)$$

$$2a = 4$$

$$a = 2$$

$$\therefore y = 2(x-1)(x-2)$$

$$= 2(x^2 - 3x + 2)$$

$$= 2x^2 - 6x + 4 \text{ Q.E.D.}$$

$$(iii) \text{area} = \int_0^2 (4 - x^2 - (2x^2 - 6x + 4)) dx$$

$$= \int_0^2 (-3x^2 + 6x) dx$$

$$= [-x^3 + 3x^2]_0^2$$

$$= -8 + 12 - (-0 + 0)$$

$$= 4$$

So 4 units²

$$(b) \int_0^1 \frac{e^x}{1+e^x} dx = \log_e c$$

$$\int_0^1 \frac{e^x}{1+e^x} dx = [\ln(1+e^x)]_0^1$$

$$= \ln(1+e) - \ln(1+e^0)$$

$$= \ln(1+e) - \ln 2$$

$$= \ln \frac{1+e}{2}$$

$$\therefore c = \frac{1+e}{2}$$

$$(c) (2k-3)x^2 + (k+1)x - 1 = 0$$

$$(i) \text{Linear if } 2k-3=0$$

$$\therefore k = \frac{3}{2}$$

$$(ii) \text{quad. w. 2 roots if } k \neq \frac{3}{2} \text{ and } \Delta > 0$$

$$\text{So } (k+1)^2 - 4(2k-3)(-1) > 0$$

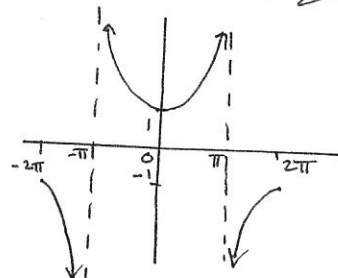
$$k^2 + 2k + 1 + 8k - 12 > 0$$

$$k^2 + 10k - 11 > 0$$

$$(k+11)(k-1) > 0$$

$$\therefore k > 1 \text{ (} k \neq \frac{3}{2} \text{)} \text{ or } k < -11$$

$$9a) i) y = \sec \frac{x}{2} \quad -2\pi \leq x \leq 2\pi$$



$$(ii) V = 2\pi \int_0^{\frac{\pi}{2}} (\sec \frac{x}{2})^2 dx$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$$

$$= 2\pi [2 \tan \frac{x}{2}]_0^{\frac{\pi}{2}}$$

$$= 2\pi [2 \tan \frac{\pi}{4} - 2 \tan 0]$$

$$= 2\pi [2 - 0]$$

$$= 4\pi \text{ so } 4\pi \text{ units}^3$$

$$(b) f'(x) = e^{-2x}$$

$$(i) \text{increasing } \because e^{-2x} > 0$$

for all real x, so grad pos for all real x

$$(ii) (0, 4.5) \text{ on curve show } f(x) = 5 - \frac{e^{-2x}}{2}$$

$$f'(x) = e^{-2x}$$

$$f(x) = -\frac{1}{2} e^{-2x} + C$$

$$4.5 = -\frac{1}{2} e^0 + C$$

$$4.5 = -\frac{1}{2} + C$$

$$C = 5$$

$$\therefore f(x) = -\frac{1}{2} e^{-2x} + 5 \text{ Q.E.D.}$$

$$(iii) \lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} 5 - \frac{e^{-2x}}{2}$$

$$= 5 - 0$$

$$= 5 \quad (\because \lim_{x \rightarrow \infty} e^{-2x} = 0)$$

9(iv) x-int: $f(x) = 0$

$$\frac{e^{-2x}}{2} = 5$$

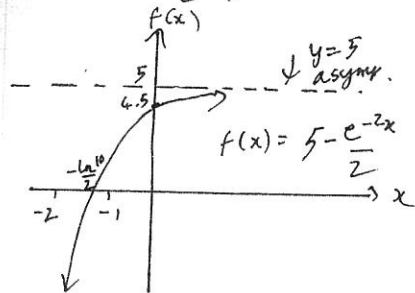
$$e^{-2x} = 10$$

$$-2x = \ln 10$$

$$x = -\frac{\ln 10}{2}$$

$$\text{So } (-\frac{\ln 10}{2}, 0)$$

(v)



10(a)(i) $m_{PA} = \frac{y-2}{x-4}$ $m_{PB} = \frac{y+8}{x+2}$

$$PA \perp PB \therefore \frac{y-2}{x-4} \times \frac{y+8}{x+2} = -1$$

$$\therefore (y-2)(y+8) = -(x-4)(x+2)$$

$$y^2 + 6y - 16 = -(x^2 - 2x - 8)$$

$$x^2 - 2x - 8 + y^2 + 6y - 16 = 0$$

$$x^2 - 2x + 1 + y^2 + 6y + 9 = 8 + 16 + 16$$

$$(x-1)^2 + (y+3)^2 = 34$$

(ii) circle centre $(1, -3)$ radius $\sqrt{34}$ u.

(b) (i) $BC = \sqrt{x^2 + 9}$ (Pythag)

(ii) $T = \frac{P}{S} \therefore \text{rowing time} = \frac{\sqrt{x^2 + 9}}{4}$

$$\text{walking time} = \frac{6-x}{5}$$

$$\therefore \text{total time } T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6-x}{5} \quad \text{QED}$$

$$\text{noting } 0 \leq x \leq 6$$

(iii) min $T(x)$

$$T(x) = \frac{(x^2 + 9)^{\frac{1}{2}}}{4} + \frac{6-x}{5}$$

$$T'(x) = \frac{\frac{1}{2}(x^2 + 9)^{-\frac{1}{2}} \times 2x}{4} - \frac{1}{5}$$

$$= \frac{xc}{4\sqrt{x^2 + 9}} - \frac{1}{5}$$

$$\text{for max/min } T'(x) = 0$$

$$\text{So } \frac{x}{4\sqrt{x^2 + 9}} = \frac{1}{5}$$

$$\therefore \frac{x}{\sqrt{x^2 + 9}} = \frac{4}{5}$$

$$\frac{x^2}{x^2 + 9} = \frac{16}{25}$$

$$25x^2 = 16x^2 + 9 \times 16$$

$$9x^2 = 9 \times 16$$

$$x^2 = 16$$

$$x = 4 \quad (x \geq 0)$$

So $x = 4$ is poss max/min

10(b)(iii)

(iii) $T'(4^+) > 0, T'(4^-) < 0$.



So min ~~at~~ at $x = 4$.

$$T(4) = \frac{\sqrt{4^2 + 9}}{4} + \frac{6-4}{5}$$

$$= \frac{5}{4} + \frac{2}{5}$$

$$= \frac{33}{20}$$

So $\frac{33}{20}$ hours.

= 1h 39min.